VIT BHOPAL	Term End Examinations – Fall Semester 2025-26 - December 2025	
	Course code : **EEE1001**	Slot : **B21+E14+E22**
	Course Title : **Electric Circuits and Systems**	Time: **180 Minutes**
	Date/ Session : **21 Jan 2025 / Session I**	Max. Marks: **100**

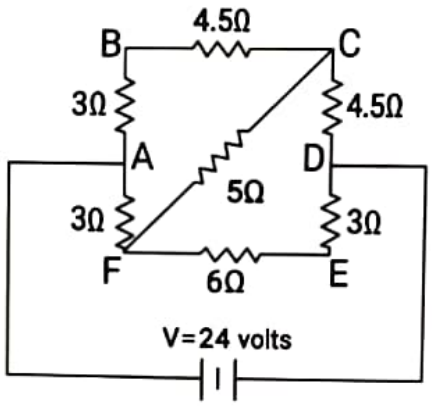
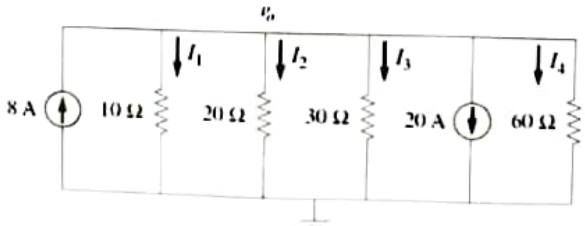
KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE

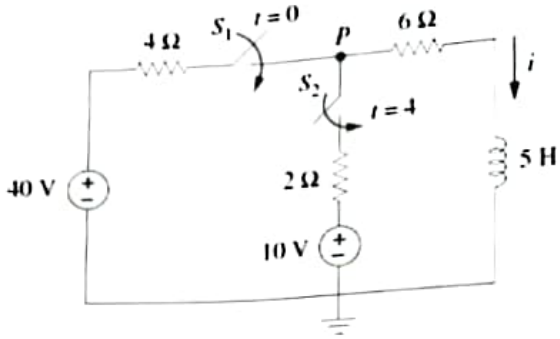
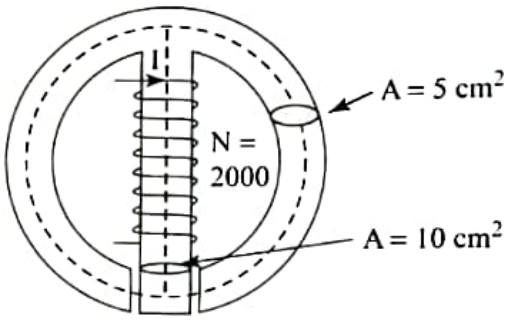
General Instructions if any:

1. **Non-Programmable calculator is permitted: YES**
2. **Reference tables are permitted: NO**

PART – A

Answer ANY FIVE Questions (5 X 12 = 60 Marks)

S. No.		Questions	Marks
1	(a)	Determine the current flowing through the branch FC in Figure 1 using Thevenin's theorem  Figure 1	12
		(OR)	
	(b)	Determine the branch currents (I_1 and I_2) and the voltage v_o for the circuit given in Figure 2  Figure 2	12
2	(a)	A resistance of 60Ω, inductance of 30 mH, and Capacitance of 5μF are connected in series across a 220V, 50Hz AC Supply. Find the following: (i) Impedance of circuit (ii) Current (iii) Power factor (iv) Voltage drop across resistor, inductor, and capacitor	12

	(b)	At t=0, switch 1 in Figure 3 is closed, and switch 2 is closed [illegible] later. Find i(t) for t>0. Calculate i for t = 2s and t = 5s  Figure 3	12
3	(a)	A parallel magnetic circuit with 2000 turns on its central limb is shown in Figure 4. The air gaps are 2 mm each. The mean diameter of the circular magnetic path is 20 cm. The cross-sectional area of the central limb is 10 cm^2, while the cross-sectional area of the outer limbs is 5 cm^2. Calculate the current, I, which must flow through the coil to produce a flux of 1.1 mWb in the central limb. The magnetic field strength, H corresponding to B = 1.1 Wb/m^2 is 650 AT/m.  Figure 4	12
		(OR)	
	(b)	With a neat and clean diagram, explain the construction, operation, and applications of a DC motor.	12
4	(a)	Discuss the operation of a Half-Wave Rectifier with diagrams and graphs as required.	8
	(b)	The input to a bridge rectifier is provided through a step-down transformer with a turn ratio of 10:1. The supply voltage is 230 V at 50 Hz. The load resistance is 1.2 kΩ, the secondary winding resistance of the transformer is 4 Ω, and the diode forward resistance is 2 Ω. Calculate the DC Power output of the rectifier.	4
		(OR)	
	(c)	Discuss the working of an n-MOSFET both as a depletion MOSFET and an enhancement MOSFET with a neat and clean diagram.	12
5	(a)	Determine the Boolean expression of Q in Figure 5.	4

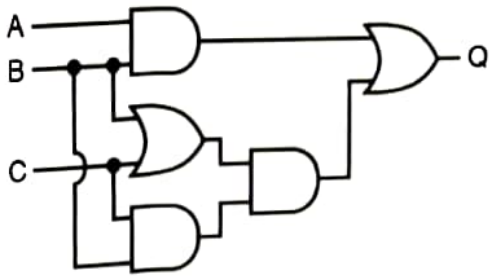


Figure 5

	(b)	Draw the Circuit of the S-R flip-flop using NAND gates and determine the following: truth table, excitation table, and the characteristic equation	8

(OR)

	(c)	Explain the Full Adder with the necessary truth table, Logic diagram, and obtain the expression for SUM and CARRY.	6
	(d)	Simplify the Boolean expressions to the minimum number of literals. (a) (A + B) (B + C') (A + C') (b) AC + (AB)' + AB'C (A + C)	6
6	(a)	For the circuit in Figure 6, use the superposition theorem to find the current i.	12

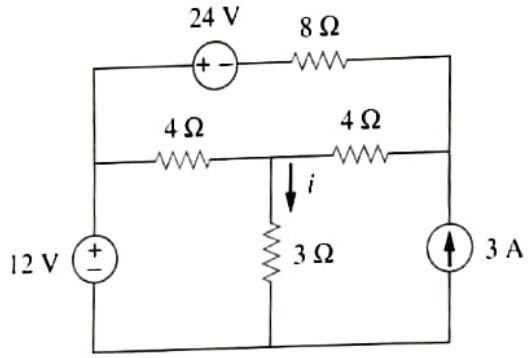


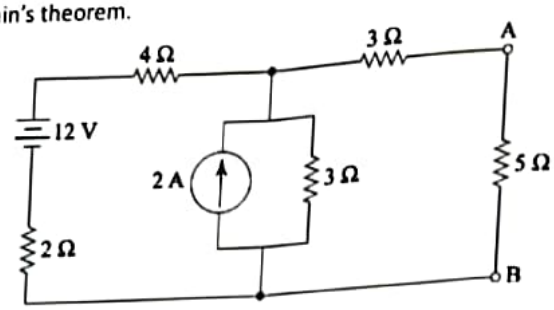
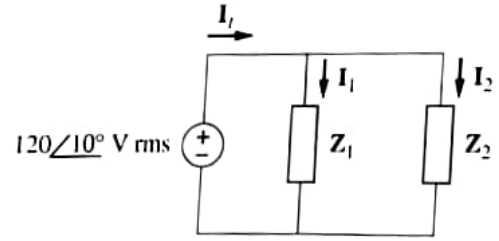
Figure 6

(OR)

	(b)	Compare the electric and magnetic circuits. A magnetic circuit has its winding on its central limb. The cross-sectional area of the central limb is 10 cm^2, whereas the cross-sectional area of the outer limbs is 5 cm^2. The effective length of the central limb is 16 cm, and that of the outer limbs is 25 cm. Calculate the current required to flow through the winding, which has 1000 turns to produce a flux of 1.2 mWb in the central limb. Assume that for a flux density of 1.2 Wb/m^2, the magnetizing force required is 750 AT/m. Draw the equivalent electric circuit also.	12

PART – B

Answer ANY FIVE Questions (40 Marks)

7	Determine the current flowing through the 5 Ω resistor in the circuit shown in Figure 7 using Thevenin's theorem.  Figure 7	8
8	An inductive coil having negligible resistance and 0.1 Henry is connected across a 200 V, 50 Hz supply. Find the following parameters: (i) Inductive reactance, (ii) RMS value of current, (iii) Power, (iv) Power factor	8
9	Explain the Principle of operation and working of the single-phase induction motor.	8
10	Explain the operation of a PN junction diode. How is it different from a Zener diode? Draw the V-I characteristics of a PN junction diode.	8
11	(i) Convert the $(100110.01011)_2$ to an Octal Number (ii) Convert $(FED.ABE)_{16}$ to a Decimal number (iii) Convert $(ABCD)_{16}$ to its equivalent octal and binary (iv) Determine the value of base x if $(212)_x = (158)_8$	8
12	The circuit in Figure 8 has the following parameters $Z_1 = 60\angle -30°$, and $Z_2 = 40\angle 45°$. Determine the power and the power factor of the circuit.  Figure 8	8

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